ORANGE Report

SAMPLE
Demo Panel

PRIMARY TUMOR	PURITY	PLOIDY	FIT METHOD	QC
NOT SPECIFIED	70% (59%-100%)	2.06 (1.92-2.32)	NORMAL	PASS

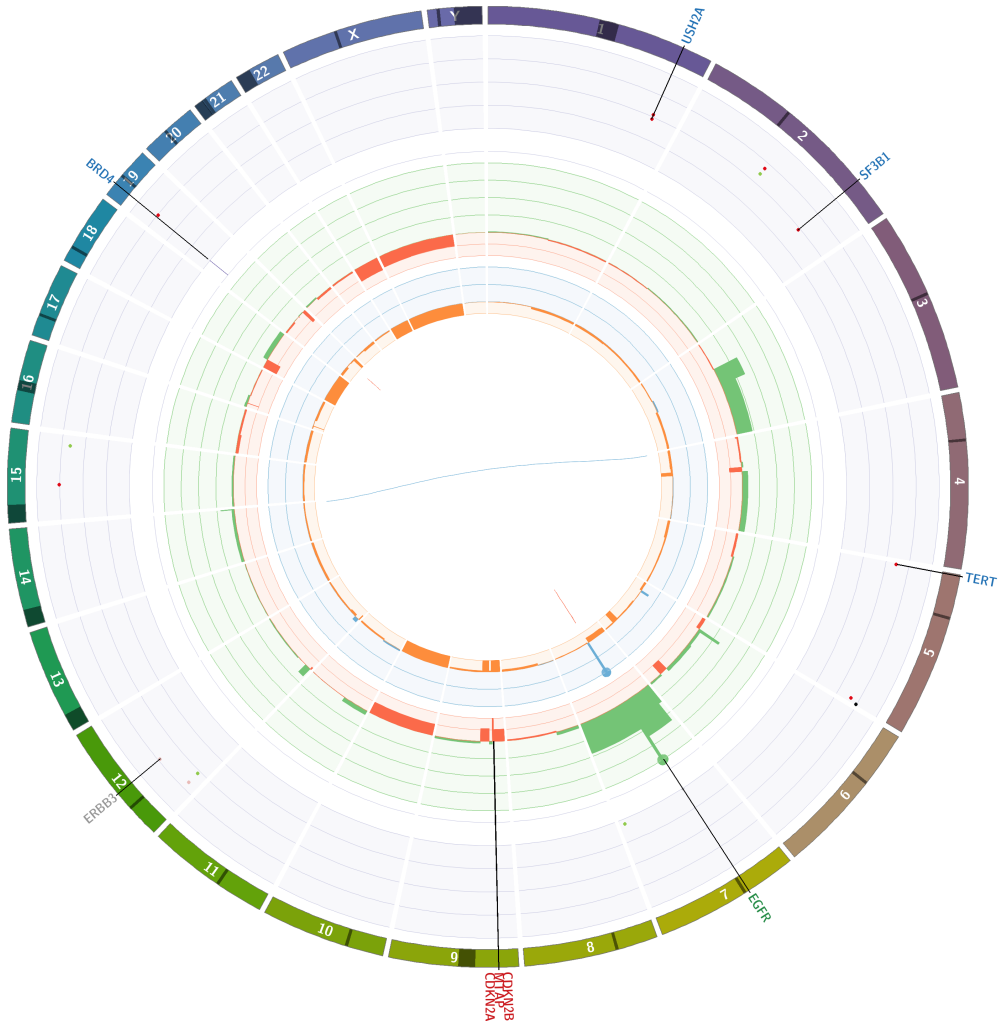
PIPELINE VERSION	GENOME VERSION	SEQUENCING TYPE	PIPELINE	SAMPLES	DATE ANALYSED
OA_V3_0	V37	ILLUMINA	PANEL: MSK-FLEX	TUMOR / RNA	2026-06-16

Driver Summary

Somatic variant:	7 (BRD4, ERBB3, SF3B1, TERT, USH2A)
Somatic copy number:	8 (CDKN2A, CDKN2B, EGFR, MTAP)
Somatic disruption:	None
Fusion drivers:	2 (@IGH::NSD2, EGFR::EGFR)
Whole genome duplicated:	No

Genome Wide Biomarkers

Microsatellite indels per Mb:	0.0 (Stable)
Tumor mutations per Mb:	3.7 (Low)
Tumor mutational load:	74 (Low)
LOH proportion:	3%
Number of SVs:	3



Somatic Findings

Small Variants (7)

GENE	POSITION	HGVS	AF	DEPTH	COPIES	HOTSPOT	BIALL	CLONAL	SOMATIC	DRIVER
BRD4	19:15353992	c.2870_2887delCCCTGCCGCCCCCACCCC [p.Pro957_Pro962del]	0.02	181	0.1 of 1.9	No	2%	0%	HIGH	100%
TERT	5:1295228	c.-124C>T	0.42	310	1.1 of 1.8	Yes	2%	100%	HIGH	100%
USH2A	1:216256859	c.5237C>T [p.Thr1746Ile]	0.33	258	0.9 of 1.9	No	2%	100%	HIGH	100%
SF3B1	2:198267698	c.1781G>A [p.Arg594Gln]	0.34	325	1.0 of 2.1	No	2%	100%	HIGH	38%
ERBB3	12:56493973	c.3145A>C [p.Ser1049Arg]	0.51	453	1.5 of 2.1	No	11%	100%	LOW	24%
MGA	15:41988683	c.1475C>T [p.Ser492Phe]	0.54	296	1.6 of 2.1	No	22%	100%	LOW	8%
FANCI	15:89849393	c.3505A>G [p.Met1169Val]	0.47	624	1.4 of 2.1	No	3%	100%	MEDIUM	0%

Amplifications and Deletions (8)

LOCATION	GENE	TYPE	RANGE	MIN CN	MAX CN	REL CN	ARM CN	TPM	PERCENTILE	FOLD CHANGE	DRIVER
7p11.2	EGFR	PARTIAL_AMP	PARTIAL	4.4	66.8	2.1	8.8	371.0	0.98	11.1	HIGH
9p21.3	CDKN2A (alt)	DEL	PARTIAL	0.0	0.0	-0.1	1.0	NA	NA	NA	HIGH
9p21.3	CDKN2A	DEL	PARTIAL	0.0	0.0	-0.1	1.0	NA	NA	NA	HIGH
9p21.3	CDKN2B	DEL	PARTIAL	0.0	0.0	-0.1	1.0	NA	NA	NA	HIGH
9p21.3	MTAP	DEL	PARTIAL	0.0	1.2	-0.2	1.0	NA	NA	NA	HIGH
10q24.32	LDB1	HET_DEL	PARTIAL	1.0	1.0	0.5	1.0	NA	NA	NA	LOW
10q25.2	SHOC2	HET_DEL	PARTIAL	1.0	1.0	0.5	1.0	NA	NA	NA	LOW
22q13.1	RAC2	HET_DEL	PARTIAL	0.9	0.9	0.4	0.9	NA	NA	NA	LOW

Fusions (2)

FUSION	JUNCTIONS	JCN	PHASING	TYPE	DRIVER
@IGH::NSD2	IG (@IGH) - Exon 6 (ENST00000482415)	2.0	Inframe	ENHANCER_KNOWN_PAIR	HIGH
EGFR::EGFR	Exon 1 (ENST00000275493) - Exon 8 (ENST00000275493)	12.6	Inframe	EXON_DEL_DUP	HIGH

Disruptions (0)

NONE

Arm Copy Number Abberations

CHROMOSOME	ARM	TYPE	CN	REL CN	DRIVER
chr3	Q	GAIN	2.9	1.4	HIGH
chr7	P	GAIN	4.4	4.3	HIGH
chr7	Q	GAIN	3.8	1.9	HIGH
chr9	P	LOSS	1.0	0.5	HIGH

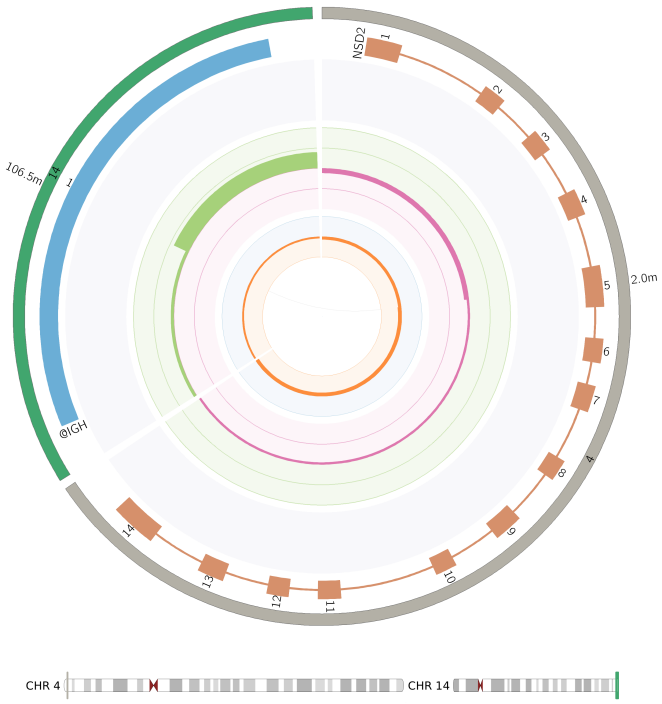
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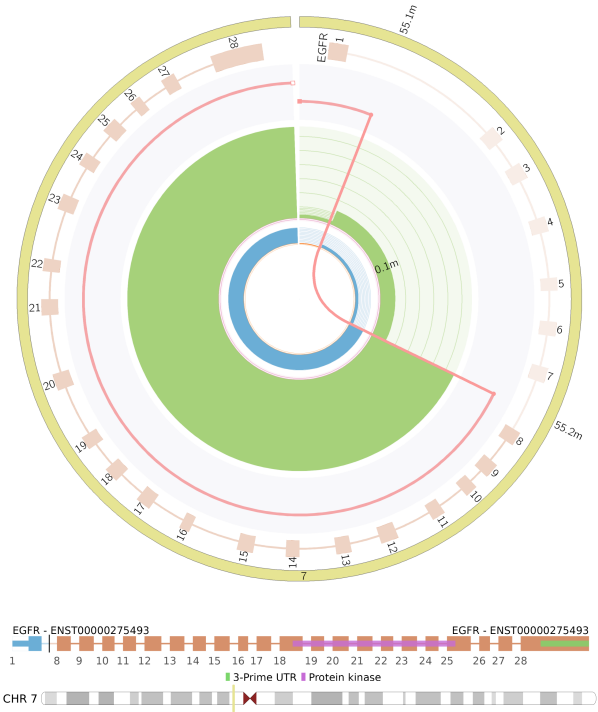
Arm Copy Number Abberations

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CHROMOSOME	ARM	TYPE	CN	REL CN	DRIVER
chr10	P	LOSS	1.0	0.5	HIGH
chr10	Q	LOSS	1.0	0.5	HIGH
chr18	P	LOSS	0.7	0.4	HIGH
chr22	Q	LOSS	0.9	0.4	HIGH

Structural Driver Plots (2)





Immunology

HLA Class I Alleles

ALLELE	QC STATUS	REF FRAGS	TUMOR FRAGS	RNA FRAGS	TUMOR CN	SOMATIC MUTATIONS
A*01:01	PASS	NA	5033	0	1.2	NONE
A*24:02	PASS	NA	4587	0	1.0	NONE
B*08:01	PASS	NA	4792	0	1.2	NONE
B*35:02	PASS	NA	4170	0	1.0	NONE
C*04:01	PASS	NA	4400	0	1.2	NONE
C*07:01	PASS	NA	5083	0	1.0	NONE

RNA Findings

QC

QC	TOTAL FRAGMENTS	DUPLICATE RATE	SPLICED RATE	UNSPLICED RATE	ALT-SLICED RATE	CHIMERIC RATE
PASS	7396614	88%	66%	5%	22%	12%

High Gene Expression (1)

GENE	TPM	PERCENTILE	FOLD CHANGE
ALK	24.3	0.99	30.0

Fusions Detected In RNA, Not In DNA (4)

FUSION	TYPE	SV TYPE	JUNC START	JUNC END	SUPPORT	COHORT FREQ
PAX8::PPARG	KNOWN_PAIR	BND	KNOWN	KNOWN	5/26	0
TFG::NTRK1	KNOWN_PAIR	BND	KNOWN	KNOWN	3/709	0
ETV6::NTRK3	KNOWN_PAIR	BND	KNOWN	KNOWN	13/47	0
TMPRSS2::ERG	KNOWN_PAIR	DEL	KNOWN	KNOWN	1/1	11

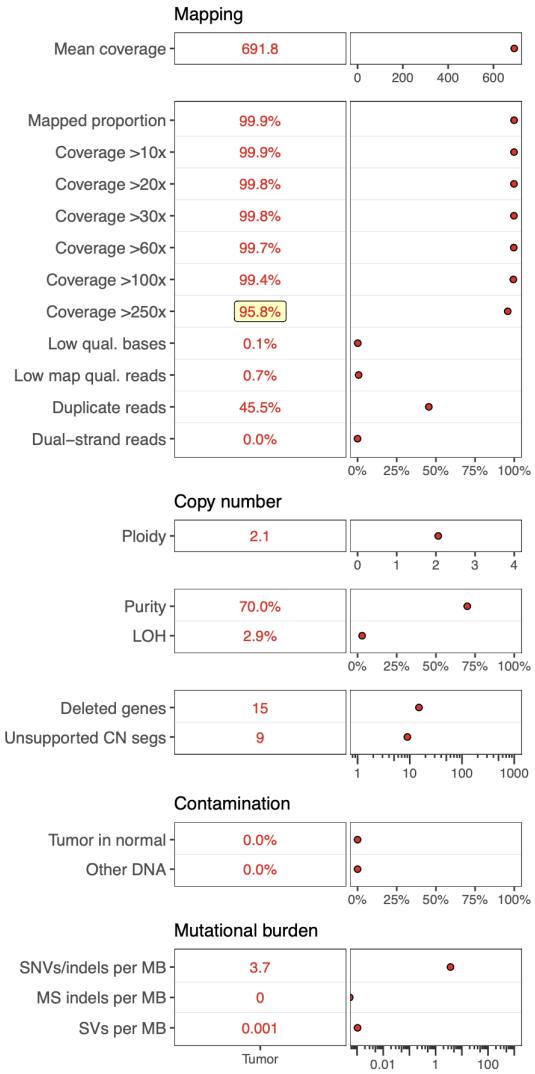
Novel Splice Junctions (1)

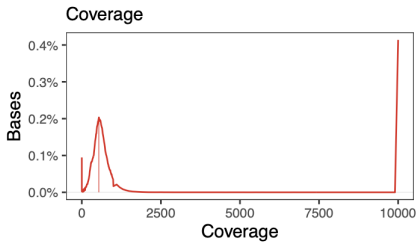
GENE	JUNCTIONS	TYPE	JUNC START	JUNC END	SUPPORT	COHORT FREQ
EGFR	Exon 1 (7:55087058) - Exon 8 (7:55223523)	SKIPPED_EXONS	SPLICE_JUNC	SPLICE_JUNC	3723/8512	15

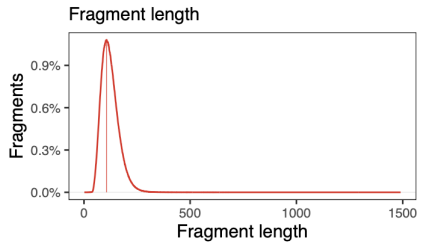
Quality Control

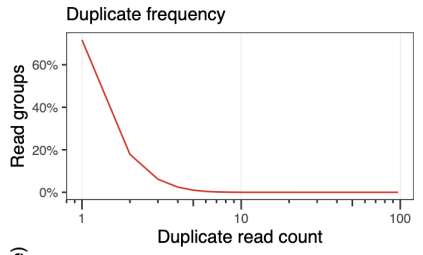
Sample_16243969–DW: WARN

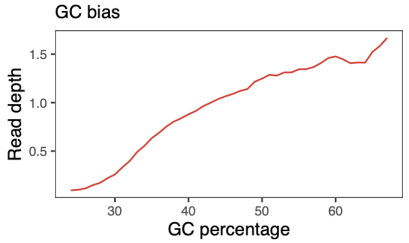
Tumor – WARN: Coverage >250x (<99%)

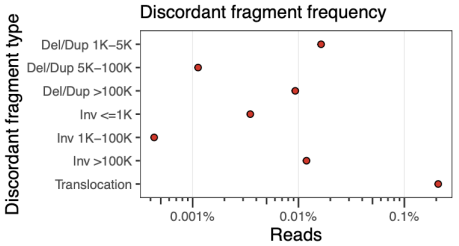


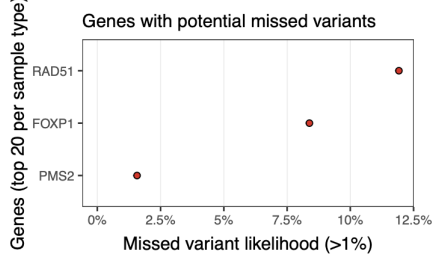


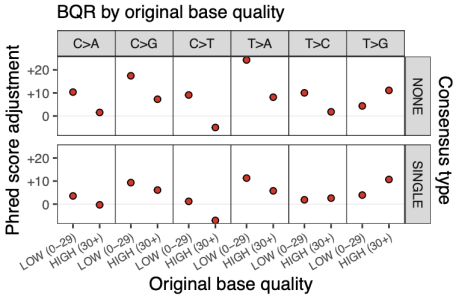


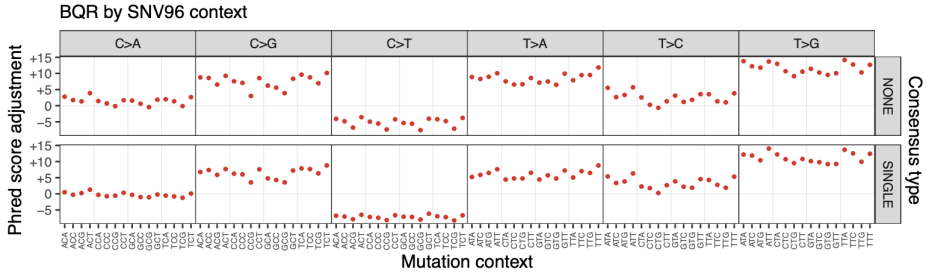


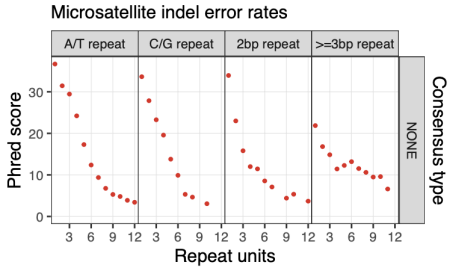


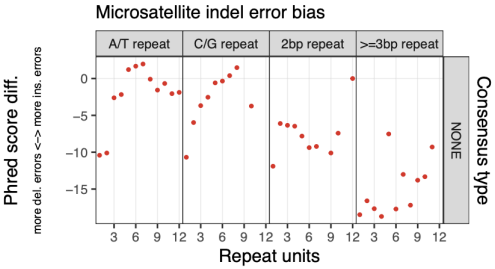


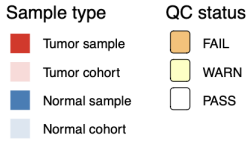












Purity and Ploidy



